

(By defn. of matrix multiplication)

$$= \begin{bmatrix} 1 \cdot 3 + 2 \cdot (-1) & 1 \cdot 5 + 2 \cdot 4 \\ (-2) \cdot 3 + 1 \cdot (-1) & (-2) \cdot 5 + 1 \cdot 4 \end{bmatrix} \quad (\text{Row-Vector Rule})$$

$$= \begin{bmatrix} 1 & 13 \\ -7 & -6 \end{bmatrix}$$

In the rest of this exercise set and in those to follow, you should assume that each matrix expression is defined. That is, the sizes of the matrices (and vectors) involved "match" appropriately.

3. Let $A = \begin{bmatrix} 4 & -1 \\ 5 & -2 \end{bmatrix}$. Compute $3I_2 - A$ and $(3I_2)A$.

$$3I_2 = 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \quad (\text{By defn. of scalar multiple})$$

$$3I_2 - A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 4 & -1 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} + (-1) \begin{bmatrix} 4 & -1 \\ 5 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} -4 & 1 \\ -5 & 2 \end{bmatrix} \quad (\text{By defn. of scalar multiple})$$

$$= \begin{bmatrix} -1 & 1 \\ -5 & 5 \end{bmatrix} \quad (\text{By defn. of matrix sum})$$

$$(3I_2)A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ -5 & 5 \end{bmatrix} = \begin{bmatrix} (3I_2)a_1 & (3I_2)a_2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 3 \cdot (-1) + 0 \cdot (-5) \\ 0 \cdot (-1) + 3 \cdot (-5) \end{bmatrix} \quad (\text{Row-Vector Rule}) = \begin{bmatrix} -3 \\ -15 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \end{bmatrix} = 1 \begin{bmatrix} 3 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 3 \end{bmatrix} \quad (\text{By defn. of } Ax)$$

$$= \begin{bmatrix} 3 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 15 \end{bmatrix} = \begin{bmatrix} 3 \\ 15 \end{bmatrix} \quad (\text{By defn. of scalar multiple})$$

(Wrong Calculation of $(3I_2)A$)